

Optimizing Sensor Placement for Flow Reconstruction in Urban Drainage Networks: A Digital Twin-Based Sparse Sensing Approach

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Abstract: Urban flooding triggered by intense rainfall is becoming increasingly frequent and widespread. While flood prediction and monitoring in high spatio-temporal resolution are desired, practical constraints in time, budget, and technology hinder its full implementation. How to monitor urban drainage networks and predict flow conditions under constrained resources is a major challenge. To address this, we introduced a **data-driven sparse sensing (DSS) approach**, demonstrated via a digital-twin of the Woodland catchment in Duluth, Minnesota. Specifically, we coupled EPA-SWMM with singular value decomposition and QR factorization-based sensor selection to optimize monitoring locations for system-level flow reconstruction. An ensemble of SWMM simulations, driven by diverse scenarios, provided the necessary hydraulic data to extract the reduced basis and identify informative sensor locations. Cross-event validation showed that three strategically placed sensors among 77 candidate nodes achieved a mean system-level Nash-Sutcliffe efficiency (NSE) of 0.949 across observed storm events. The QR-selected sensor sets were benchmarked against reference sensor configurations obtained from exhaustive searches and Monte Carlo random-placements. This comparison further showed that flow reconstruction based on QR-selected sensors closely tracked the exhaustive optimum while substantially outperforming

random placements. We further evaluated the framework's robustness by introducing multiplicative Gaussian noise and simulating individual sensor failures. While the model is relatively resilient to noise, the impact of sensor dropouts depends heavily on the number of sensors allocated and their specific locations. These findings establish a digital-twin-based foundation for sparse sensor placement in urban drainage networks with potential extensions toward field deployment, predictive monitoring, and real-time flood management.

Keywords: Stormwater management, storm sewers, data-driven sparse sensing, digital twin, sensor placement, flood

Key Points

- A data-driven sparse sensing framework was utilized to identify the optimal locations to monitor storm sewers and reconstruct full-system flows.
- Data collected at three optimally placed sensors among 77 nodes achieved satisfactory performance in spatial-temporal flow reconstruction.
- The model's reconstruction performance showed good robustness to uncertainty in measurements and sensor failures.

Plain Language Summary

Urban flooding from heavy rain is becoming more common, but monitoring drainage systems in detail is often too costly and complex. This study developed a new “data-driven sparse sensing” approach that uses advanced math and computer modeling to figure out where a few sensors can be placed to capture the most important information about a city's stormwater system. Using a real drainage network in Duluth, Minnesota, we ran hundreds of simulations to train the model, then tested how well it could estimate water flows during storms. We found that **just three well-placed sensors could accurately reconstruct flow patterns across the entire network. While the model handles noisy data well, its performance when sensors fail depends a lot on how many sensors are used and where they are placed.** The method offers city planners and engineers a practical way to monitor flood risks and potentially power early warning systems – without the high cost of installing sensors everywhere.

1. Introduction

Population growth triggers urbanization and expands impermeable surfaces across landscapes (Dadashpoor et al., 2019; Wong & Kerkez, 2018). Concurrently, climate change introduces considerable uncertainty in global weather patterns, potentially causing increased rainfall intensity and frequency in various regions (Donat et al., 2016; Katherine et al., 2023; Martel et al., 2021; Xu et al., 2024). Urban surface water flooding, intensified by climate change and rapid urbanization, has emerged as a critical threat to infrastructure resilience and public safety. Statistics reveal the staggering human and economic cost of floods: tens of thousands of fatalities and billions of euros in damages have been recorded globally, including Europe and Australia (S. Grimaldi et al., 2019; Munawar et al., 2022), as well as in the United States, where annual flood-related economic losses are projected to escalate from \$7–9 billion (1903–2014) to \$19 billion by 2100 (Ntelekos et al., 2010; USGCRP, 2014; National Academies, 2019). The magnitude of these impacts underscores the urgent need for more effective flood monitoring and forecasting, which informs effective flood prevention and mitigation.

Flood monitoring, however, faces significant constraints. Despite recent development in sensor technology, high-precision equipment is still expensive for wide deployment (De Groeve et al., 2015; Tao et al., 2024). Traditional gauging stations provide only localized data, limiting effective flood management (Rahman & Di, 2017). Satellite-based remote sensing, while offering broader spatial coverage, struggles with obstructions such as cloud cover, vegetation interference, and complex image analysis requirements (Hashemi-Beni et al., 2024). These limitations are exacerbated in intricate urban environments, where urban drainage systems complicate data collection, and developing regions with insufficient monitoring networks, incomplete historical records, and sensor inaccessibility during floods (Al-Suhili et al., 2019; Mehmood & Rasmy, 2020).

This challenge can be potentially addressed by deploying sensors at the most representative locations (optimized sensor placement, OSP) to maximize information gain and improve system observability within constrained budget and resources. The value of OSP in improving monitoring accuracy and operational efficiency has been validated across diverse domains such as flood forecasting, structural dynamics, and agro-hydrology (Fattoruso et al., 2015; Sahoo et al., 2019; K. Wang et al., 2020). Contemporary research on OSP can be categorized into several major methodological approaches, including deterministic model-based optimizations, probabilistic

information-theoretic and Bayesian strategies, heuristic and evolutionary optimizations, and data-driven and learning-based methods. Deterministic model-based optimizations treat the problem as exact and pick the subset of sensor locations that maximizes an algebraic measure of observability or identifiability (Krause et al., 2008). Probabilistic information-theoretic and Bayesian strategies treat the problem as random and choose sensors that minimize posterior entropy, maximize mutual information, or equivalently minimize expected posterior variance (Ercan et al., 2023; Y. Yang et al., 2022). Heuristic and evolutionary optimizations formulate OSP as a multi-objective combinatorial problem and search the discrete space with algorithms such as genetic algorithms or particle-swarm for Pareto fronts, addressing trade-offs between competing criteria like coverage, redundancy, and robustness (Hassani & Dackermann, 2023; Lin et al., 2020; C. Yang, 2021). Moreover, data-driven and learning-based methods learn a placement policy directly from data via supervised or reinforcement learning to predict or optimize sensing layouts without an explicit physical model (Liu & Yin, 2024; Z. Wang et al., 2020).

In addition to deploying sensors at the effective locations for flood monitoring, flood map representation, prediction, or forecasting at high resolution based on physics (i.e., physics-based models) and data (i.e., learning-based models) are also critical as they inform flood prevention and mitigation measures directly. Physics-based models such as 1D hydraulic models (e.g., EPA SWMM, MIKE Urban) can be used or further integrated with 2D hydrodynamic models, either through one-way coupling (e.g., SWMM + PCSWMM) or fully dynamic 1D-2D coupling (e.g., MIKE FLOOD, Infoworks ICM) (Cheng et al., 2017; Kadam & Sen, 2012; Sidek et al., 2021; Tansar et al., 2020) to predict flowrates and flood depth across the system considering a tradeoff between computational efficiency and spatial representation. Yet, their reliance on high-resolution data and computational power limits real-time forecasting, particularly during flash floods (Berkhahn et al., 2019; Bisht et al., 2016). This tension between accuracy and efficiency has driven interest in alternative approaches.

Machine learning (ML) and deep learning (DL) models (e.g., artificial neural networks (ANNs), long short-term memory networks (LSTMs), and random forests (RF)) can learn complex patterns from historical data, enabling predictions of flood depths, durations, and extents with minimal computational resources once trained, which is particularly beneficial for large urban storm sewer systems (Chang et al., 2014; Fang et al., 2021; Gude et al., 2020; Kim & Kim, 2020; Zou et al.,

2023). For instance, Guo et al. (2021) demonstrated that deep convolutional neural networks can reduce computational time by 99.5% compared to some physically based models, a significant advancement for operational forecasting. However, they face several challenges such as data scarcity, especially for rare extreme events, and the risk of overfitting, which can limit generalization to new scenarios. Additionally, the lack of physical interpretability poses a barrier, as these models may provide accurate predictions without offering insights into underlying physical processes, which is critical for decision-making on flood prevention and mitigation.

Data-driven sparse sensing (DSS), proposed by Manohar et al. (2018), provides a framework that can potentially optimize sensor placement and simultaneously reconstruct flow conditions based on downsized measurements. First, DSS employs Singular Value Decomposition (SVD), a dimensionality reduction technique, to identify the space where the signals exhibit sparse dynamics, and then applies QR factorization with column pivoting to determine the most representative data points – corresponding to the optimal sensor locations. Second, DSS builds on the theory of compressed sensing, which considers natural signals (such as flowrates and flood depths) as “sparse” or can be represented by fewer states (or parameters) in the frequency domain, meaning these signals can be effectively reconstructed with fewer measurements (Donoho, 2006; K. Zhang, Bin Mamoon, et al., 2023). Combining both techniques, DSS can identify the most representative sensor placements and use measurements taken at those locations to reconstruct signals (e.g., flowrates or flood depths in storm sewer systems). DSS balances the advantages of physics-based models and ML/DL models as it is relatively more computationally efficient and explainable given its basis on fundamental linear algebra. For example, Ohmer et al. (2022) used DSS to optimize groundwater monitoring networks, achieving a reconstruction error of 0.1 m with 94% subset reduction. Zhang et al. (2023) and Bin Mamoon et al. (2025) applied DSS to optimize sampling times and estimate stream flow and nutrient concentrations and loads in various streams across the US. Zhang et al. (2023) found that 5 measurements taken in a year (98% reduction in data) can well reconstruct streamflow in snowfall dominated regions. Bin Mamoon et al. (2025) found that as few as 20 samples in a year can accurately estimate nutrient concentrations and loads, achieving error margins of $\pm 2\%$ for NO_x and $\pm 9\%$ for total phosphorus. Despite its validated versatility and compatibility to different systems, DSS has not been used for storm sewer systems which should be more challenging due to the flashy flow regimes. This is a critical gap given that most

municipalities face severe resource constraints and an urgent need for cost-effective sensor placement optimization.

The objective of this study is to evaluate the performance and robustness of DSS in optimizing sensor placements and reconstructing flows in storm sewer systems via a case study in Duluth, Minnesota, United States. We used an EPA-SWMM model to generate simulation training data and feed it to DSS to identify the most informative sensor locations. We then used the spatially sparse measurements obtained from this subset of nodes to reconstruct full-system flow conditions. We compared the flow reconstruction performance of DSS-obtained sensor configurations against other benchmark approaches. Furthermore, we systematically evaluate the robustness of the framework in flow reconstruction under realistic conditions, including environmental noise and partial sensor failure, to assess its reliability and readiness in practical deployment. The results provided can advance our knowledge on sensor observability in storm sewer networks and offer insights for the future practice of parsimonious sewer flow monitoring.

2. Data and Methods

2.1. Data-Driven Sparse Sensing

DSS integrates SVD and QR factorization to optimize sensor placements and efficiently represent and reconstruct data. In brief, SVD can identify a reduced-dimension space onto which the signal (e.g., flowrates in the storm sewer system, organized as matrix) is projected, while QR factorization can pivot the column of the matrix – the sensor locations – that possess the maximum information. Below further illustrates the principle of the techniques.

Natural signals, including both 1-D temporal data (e.g., runoff hydrographs) or 2-D spatial data (e.g., spatial maps of flowrates) can be represented by discrete time-series $\mathbf{x}_i$. These time-series can be represented by a linear combination of appropriate basis vectors arranged into a matrix $\mathbf{\Psi} = [\psi_1, \psi_2, \dots]$, with amplitudes, $\mathbf{a}_i$, i.e.,

$$\mathbf{x}_i = \mathbf{\Psi} \mathbf{a}_i \quad (1)$$

Most of these natural signals are “sparse”, meaning that only a few coefficients in $\mathbf{a}_i$ have large values when the time-series is represented in terms of an appropriate basis. Often, a generic or universal basis, such as Fourier or wavelets, can represent the signal sparsely without prior

knowledge of the signal properties. Traditional compressed sensing relies on these universal basis sets, which are not tailored to the specific spatio-temporal structure of the system being modeled. As a result, signal sparsity in these bases may be suboptimal, leading to less efficient reconstructions or requiring more measurements to achieve a given level of accuracy. Additionally, traditional compressed sensing typically involves random or incoherent sampling strategies, which do not exploit any physical knowledge about the sensing process or the locations of maximum information gain. In engineering systems such as stormwater networks, purely random measurements are neither practical nor efficient due to the cost and space constraints associated with sensor deployment. However, with some physical understanding of the processes that generate the signal, or with access to prior data (or computational data), it is possible to obtain a basis that is tailored to a specific signal.

In this project, we propose to develop a tailored basis from a singular value decomposition (SVD) to yield the optimal least-squares approximation to the data (Figure 1). That is, given a training dataset containing computational time series $\mathbf{X} = [x_1, x_2, \dots]$, the SVD

$$\mathbf{X} = \mathbf{\Psi}\mathbf{\Sigma}\mathbf{V}^T \quad (2)$$

identifies the orthonormal temporal basis, $\mathbf{\Psi}$, $\mathbf{\Sigma}$ holds the singular values, and $\mathbf{V}^T$ comprises the right singular vectors. The first r vectors in $\mathbf{\Psi}$, i.e., $\mathbf{\Psi}_r = [\psi_1, \psi_2, \dots, \psi_r]$ represent the optimal r temporal basis functions for the dataset. Given a limited set of measurements $\mathbf{y}$ sub-sampled from the target time series $\mathbf{x}$, where $\mathbf{y} = \mathbf{C}\mathbf{x}$ and $\mathbf{C}$ is a sampling operator representing the locations at which measurements are taken, the coefficient vector $\hat{\mathbf{a}}$ can be estimated from measurements as

$$\hat{\mathbf{a}} = (\mathbf{C}\mathbf{\Psi}_r)^\dagger \mathbf{y} \quad (3)$$

to yield the estimate for the target time series

$$\hat{\mathbf{x}} = \mathbf{\Psi}_r \hat{\mathbf{a}} \quad (4)$$

Importantly, the locations where stormwater runoff measurements are recorded can be optimized to best sample the r basis modes in $\mathbf{\Psi}_r$. These optimal sampling points can be obtained using QR factorization with column pivoting (Manohar et al., 2018),

$$\mathbf{\Psi}_r^T \mathbf{C}^T = \mathbf{Q}\mathbf{R} \quad (5)$$

In summary, the DSS framework obtains a tailored coordinate system or basis (ψ_r) via a SVD from a training dataset generated by the EPA-SWMM model. Furthermore, the most informative locations to collect samples (C) are obtained using QR factorization on the tailored basis. If a small number of measurements can be taken at these optimal sampling locations in a stormwater network, then the full map of runoff flowrates can be reconstructed from a basis trained on simulation data from a different scenario.

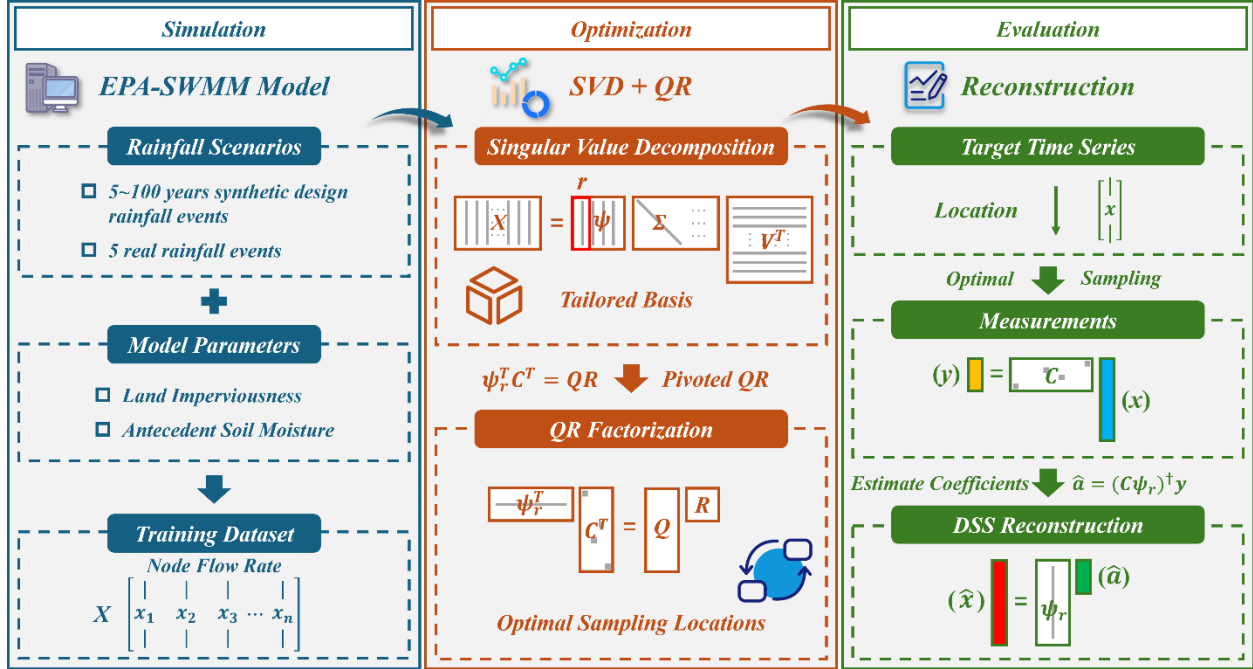


Figure 1. Methodologies and roadmaps of the data-driven sparse sensing (DSS) framework for sensor placement and urban flood reconstruction.

2.2. Case Study

This study focuses on the **Woodland catchment in Duluth, Minnesota, United States** (Figure 2), which spans 133.53 hectares (1.34 km²) and serves a population of approximately 6,244 residents. Duluth, Minnesota is a city nestled at the westernmost tip of Lake Superior, unique for its steep hills and abundance of waterways. However, this unique location coupled with changing climatic conditions has made the city increasingly vulnerable to the devastating effects of flooding. In recent years, Duluth has experienced an increase in the frequency and intensity of extreme precipitation events, posing significant challenges to its infrastructure, environment, and the well-being of its residents (City of Duluth, Minnesota, 2023). Furthermore, local waterways are highly

sensitive to stormwater runoff, as they consist of cold-water trout streams feeding directly into Lake Superior, a globally significant and sensitive aquatic ecosystem. Therefore, it is critical to monitor the flow conditions with an optimized sensor placement strategy in these storm sewers.

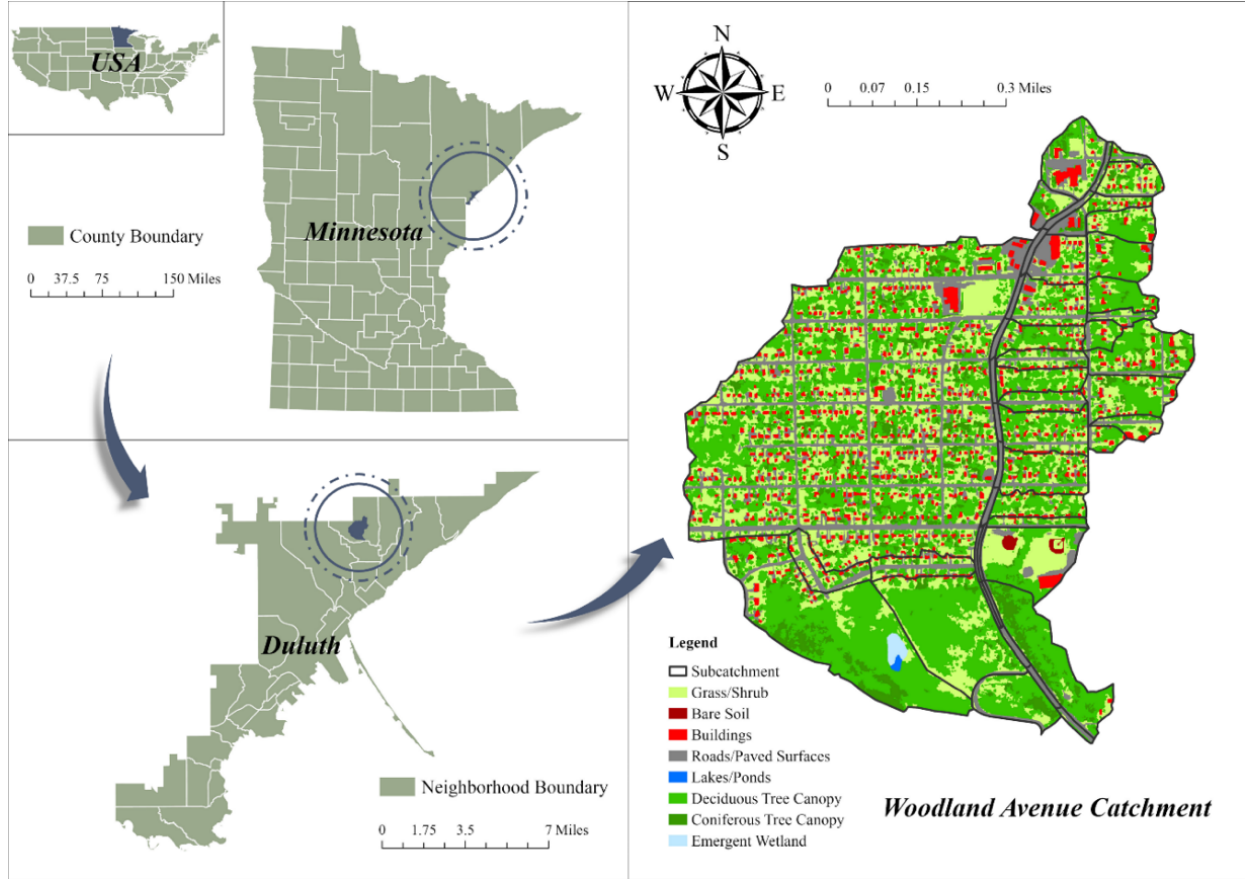


Figure 2. The location, land use and the subcatchments of the case study site.

In this catchment, land cover is dominated by permeable areas (74.74%), primarily deciduous tree canopy (40.27%), grass/shrub (26.57%) and coniferous tree canopy (7.42%). Impervious surfaces (25.26%) include roads/paved surfaces (17.59%) and buildings (7.68%), which contribute significantly to surface runoff. The catchment's soils are classified as Type C (slow infiltration), further limiting groundwater recharge and amplifying surface runoff during storms.

It should be noted that the objective of this study is not to provide practical guidance to the sensor placement in this specific catchment. Instead, the goal is to present a methodology that can be potentially transferred to other systems.

2.3. SWMM Modeling

To generate the datasets used for reduced-order sensor placement and reconstruction, we used **EPA-SWMM to simulate storm-sewer flows in the catchment** under a structured ensemble of rainfall and hydrologic parameter scenarios. The modeled drainage system contains 48 subcatchments, 130 conduits, and 77 nodes. Figure S1 in the supplementary material shows the layout of the subcatchments and sewer networks represented in the model.

The training ensemble combined 10 rainfall events with 25 hydrologic parameter settings. The rainfall set included five synthetic design storms with return periods of 5, 10, 25, 50, and 100 years, together with five observed storm events. Hydrologic variability in the catchment was introduced through five imperviousness levels (5%, 15%, 25%, 35%, and 45%) and five Horton minimum infiltration rates (0.5, 1.0, 1.5, 2.0, and 2.5 in/hr), giving 250 training scenarios in total. For each training scenario, the simulated flow field was represented by the complete time series of all 77 monitored nodes. This full spatiotemporal matrix was normalized via global min-max normalization, and it was used to construct the reduced basis and identify the DSS sensor sets. The method of normalization used can impact the results; and a comparison between global min-max normalization versus Z-score normalization can be found in Text S2 in the supplementary material. Independent testing was then performed on the six events that were excluded from basis construction, including the remaining five observed storm events and one separate 200-year design storm. The same 77-node spatiotemporal representation was used to reconstruct network flows and evaluate performance on these testing events.

Due to lack of monitoring data, the model was parameterized based on our knowledge of the catchment and sewer networks without validation with field measurements. The parameters related to sewer hydraulics were estimated based on site surveys and information extracted from the sewer system database. Some hydrologic and surface-routing parameters, e.g., surface roughness, depression storage, and overland flow width, were assumed to be uniform at the catchment scale. Despite the lack of validation, the model is considered reliable, having been previously utilized by the City of Duluth and St. Louis County to estimate the hydraulic loads into a downstream green stormwater infrastructure at its design phase. Consequently, the model was used here as a physically grounded scenario generator that preserves the real network layout and dominant hydraulic routing behavior, rather than as a fully calibrated operational digital twin. To test the

robustness of our DSS model to the parameterization in SWMM, we conducted an additional parameter perturbation experiment, which is introduced in the next section.

2.4. Validation and Performance Evaluation

To evaluate the performance of the DSS framework in identifying the most representative nodes for measurement and reconstructing flowrates in the sewer network with imperfect, sparse measurements, we designed a multi-dimensional evaluation experiment (Figure 3).

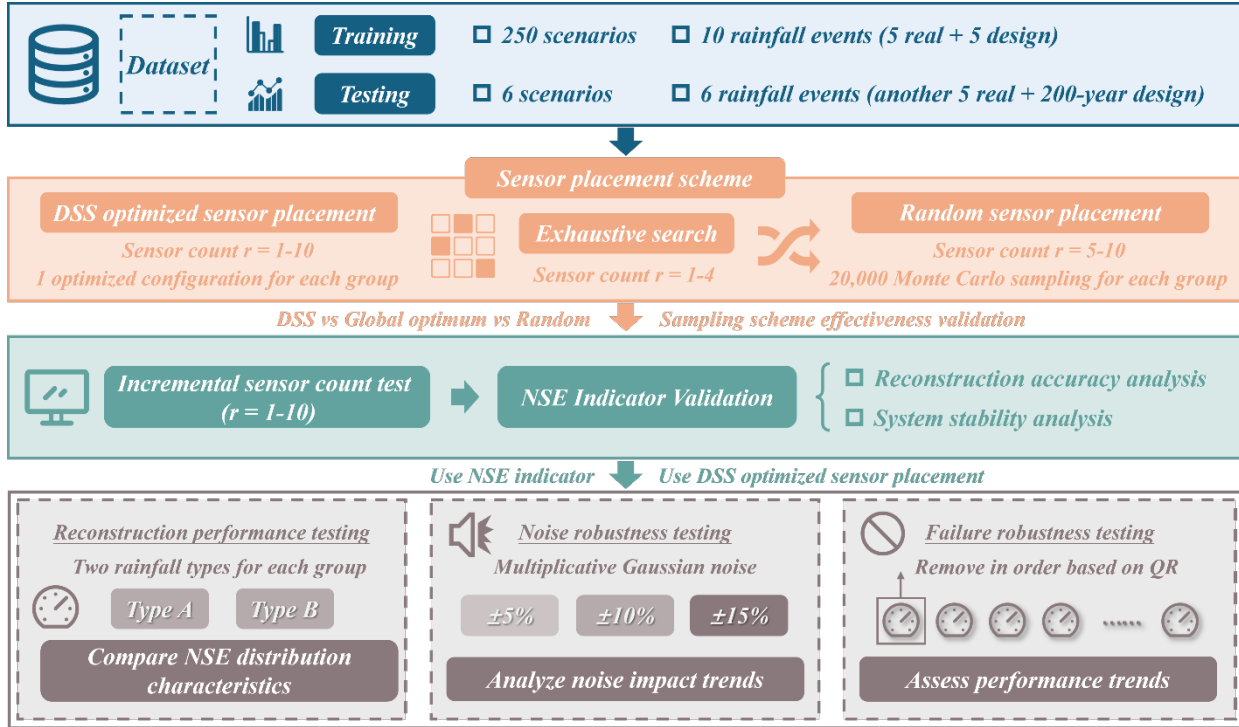


Figure 3. Methodologies and roadmaps of the validation and performance evaluation.

The validation of the reconstructed flows for all the comparison experiments was conducted using the Nash-Sutcliffe Efficiency (NSE) metric, which quantifies the agreement between the sparse sensing-reconstructed values and the SWMM-simulated observations. The NSE was calculated as:

$$NSE = 1 - \frac{\sum_{i=1}^n (Q_{SWMM,i} - Q_{DSS,i})^2}{\sum_{i=1}^n (Q_{SWMM,i} - \overline{Q_{SWMM}})^2} \quad (6)$$

where $Q_{SWMM,i}$ and $Q_{DSS,i}$ represent the SWMM-simulated and DSS-reconstructed flowrates at time step i , respectively (CFS). $\overline{Q_{SWMM}}$ is the mean of the SWMM-simulated flowrates across time steps. The closer the NSE is to 1, the higher the reconstruction accuracy. Two types of NSE values

were calculated. First, a “system-level” NSE was computed by including the flow series over all included nodes and time steps within each event. Second, a “node-level” NSE was computed separately for each included node and then summarized across nodes. The system-level NSE quantifies the flow reconstruction efficiency across the entire network, whereas the node-level NSE quantifies the flow reconstruction efficiency of each node.

In total, five different experiments were performed as explained below.

1. **Cross-event reconstruction performance testing:** We evaluated the flow reconstruction accuracy of DSS-obtained sensor configurations for both real and synthetic design rainfall events at every sensor count from $r = 1$ to $r = 10$. For this test, both system-level and node-level NSEs were calculated and compared.
2. **Validation of sampling scheme effectiveness:** We evaluated the effectiveness of DSS-obtained sensor configurations by comparing their flow reconstruction efficiency against benchmarks obtained from an exhaustive search (with sensor counts $r = 1-4$) and random placements (with sensor counts $r = 5-10$). For $r = 1-4$, all feasible combinations were exhaustively enumerated, and the global optimum was identified. For $r = 5-10$, the random benchmark was estimated by Monte Carlo sampling with 20,000 configurations per sensor count, verified to be sufficient by ensuring that the central boxplot statistics (e.g., median) had reached a stable, converged state. For this test, only system-level NSE was compared.
3. **Environmental noise robustness testing:** environmental noise is inevitable in real-world applications, particularly in urban hydrological sensing systems where sensor readings are susceptible to external interferences. To evaluate the robustness of our reconstruction framework under noisy conditions, and weigh the degradation caused by measurement noise against the performance gains of increased sensor density, we applied a multiplicative Gaussian measurement-noise model to the selected sensor signals. For each clean sensor signal y , the noisy observation was generated as:

$$y_{noise} = y(1 + e) \quad (7)$$

$$e \sim \mathcal{N}(0, \sigma^2), \sigma = \text{noise level}/3 \quad (8)$$

where ϵ follows a zero-mean Gaussian distribution. We considered three noise levels, corresponding to $\pm 5\%$, $\pm 10\%$ and $\pm 15\%$ of uncertainty. The standard deviation of the Gaussian noise term was set to one third of the prescribed noise level so that 99.7% of realizations fell within the target uncertainty range under the three-sigma rule. For each sensor count from $r = 1$ to $r = 10$, 1,000 Monte Carlo realizations were generated and the system-level NSE distributions were computed from the corresponding noisy reconstructions. For this test, only system-level NSE was compared.

4. **Sensor failure robustness testing:** sensor failures are inevitable in real-world monitoring systems due to **hardware malfunction, environmental conditions, or maintenance issues**. To understand how such failures affect the reconstruction performance, we sequentially removed one selected sensor at a time based on the order of sensor locations obtained from QR factorization and evaluated the reconstruction accuracy based on the remaining sensors with number of sensors increased from 1 to 10. Crucially, since the DSS sensor set is re-optimized for each r , this process assesses the fault tolerance of specific sensing configurations rather than merely evaluating subsets of a single higher-density design. For this test, only system-level NSE was compared.

5. **Model uncertainty robustness testing:** To test whether the main conclusions depended too strongly on the simplified baseline parameterization, we constructed an additional independent SWMM perturbed ensemble and compared the flow reconstruction and optimal sensor configurations obtained from the baseline versus perturbed ensembles. More specifically, the baseline network topology and structure were preserved, but spatially distributed perturbations were applied to several hydraulic and hydrologic parameters that were homogenized or weakly varying in the baseline SWMM model. These perturbations included variations in subcatchment width, Horton infiltration parameters, conduit roughness, as well as global and zoned (upstream versus downstream) changes in surface roughness and depression storage. In total, 9 additional SWMM uncertainty variants were created and tested (10 including the baseline case). More details of these uncertainty variants can be found in the Text S1 and Table S1 in the supplementary material. For each case, the full training ensemble was regenerated using the same ten training rainfall events and the same 25 hydrologic parameter combinations (250 scenarios for each variant). The DSS approach was applied to each training ensemble, and the optimal sensor placements and system-level flow reconstructions obtained from each training ensemble were compared against

each other to evaluate the robustness of the methodology to model parameterization. For this test, the robustness was evaluated based on both the overlap between the selected sensor sets and the system-level NSE.

3. Results

3.1. System-Level Reconstruction Accuracy

System-level reconstruction performance improved consistently with increasing sensor count. For the five observed rainfall events, the mean system-level NSE increased from 0.896 at $r = 1$ to 0.993 at $r = 10$. The 200-year design event showed the same overall trend; its system-level NSE increased from 0.842 at $r = 1$ to 0.994 at $r = 10$ (Fig. 4a). This increase in reconstruction performance can be clearly seen from the scatter-density plots shown in Fig. 4b1–b3. With $r = 1$, DSS slightly underestimated the peak flowrates (Fig. 4b1); however, that underestimation was resolved with more sensors included (Fig. 4b2–b3).

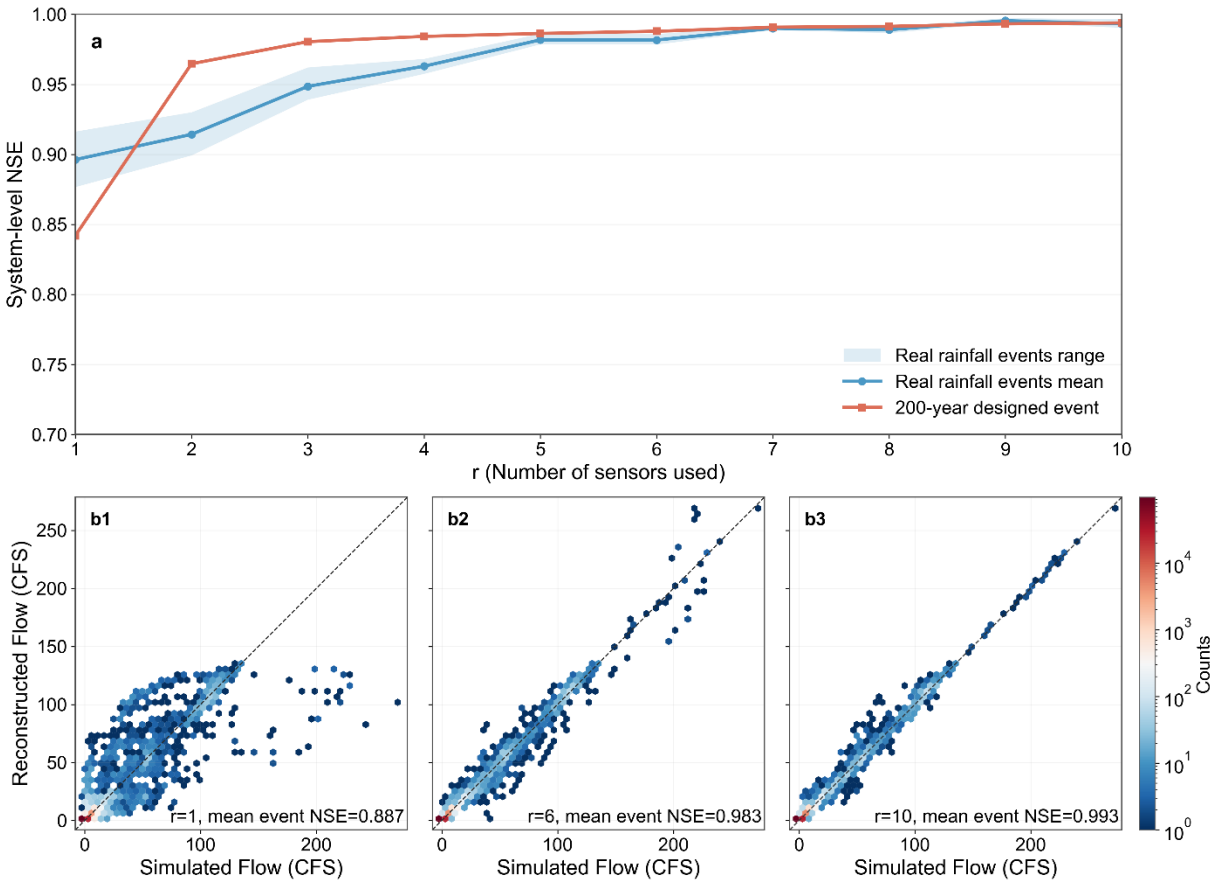


Figure 4. System-level flow reconstruction performance represented by system-level NSE values. Panel a shows the mean and range of system-level NSE for the five real rainfall events together with the 200-year design event against sensor number r . Panels b1–b3 compare simulated and reconstructed flows for representative cases $r=1$, $r=6$ and $r=10$ using hexagonal binning density plots.

A moderate number of strategically placed sensors was sufficient to recover the dominant system dynamics, whereas additional sensors mainly provided incremental refinement. For the five observed rainfall events, the mean system-level NSE became nearly stationary (> 0.982) with $r = 5$. Adding more sensors reduced the spread across events. The 200-year design event showed stronger sensitivity to sensor count; Its system-level NSE became almost stationary (> 0.914) with $r = 2$ (Fig. 4a). This indicates that while extreme-event reconstruction was more vulnerable under highly sparse monitoring, the DSS-selected sensors efficiently recovered the system-wide hydraulic response once a small number of informative locations were available.

Hydrograph comparisons confirmed that adding more sensors did not merely inflate overall performance metrics, but physically improved the capture of critical flow features, such as peak timing and transient flow behaviors. Fig. 5 compares the reconstructed versus SWMM-simulated hydrographs for three storm events under three representative sensor-count settings ($r = 1, 6, 10$ corresponding to low, intermediate, and higher levels of sensing within the candidate set). Across all observed events, increasing the sensor count generally reduced discrepancies in peak flow, timing, and volume. For instance, increasing r from 1 to 10 reduced peak-flow errors from 0.95% to 0.54% for the June 2nd, 2024 event (Fig. 5a) and from 10.92% to 2.14% for the June 11th, 2024 event (Fig. 5b). The most significant gains occurred in the 200-year design event, where peak-flow error dropped sharply from 27.14% ($r = 1$) to 0.18% ($r = 10$), and time-to-peak errors were eliminated entirely (Fig. 5c). These quantitative gains were mirrored visually: as r increased, hydrograph reconstructions transitioned from capturing general trends to faithfully recovering peak magnitudes and recession behaviors, evidenced by the marked narrowing of shadow-line envelopes.

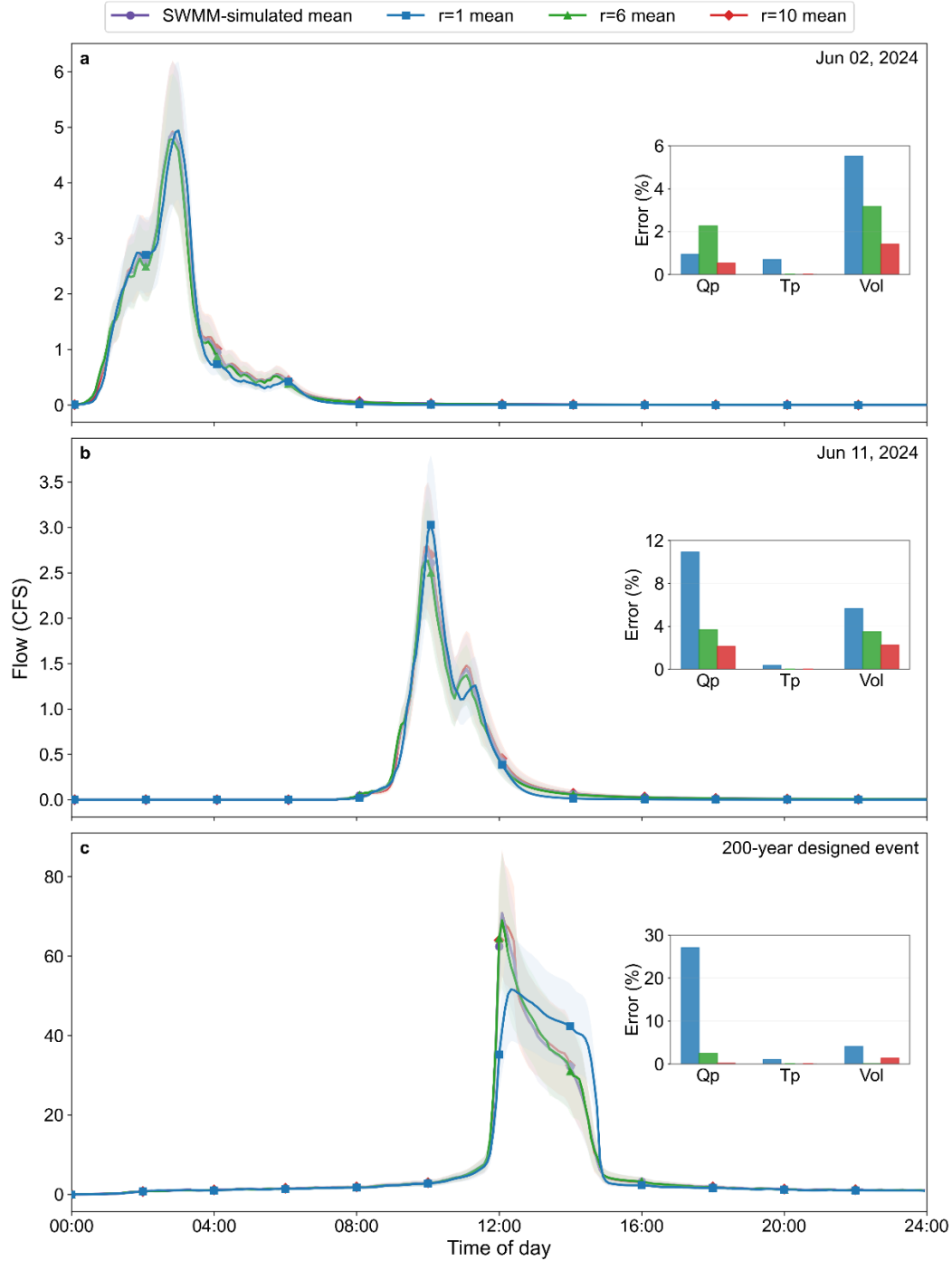


Figure 5. Hydrograph comparisons under three representative sensor-count settings. Simulated and reconstructed hydrographs are shown for representative (a–b) observed and (c) designed rainfall events at $r = 1$, $r = 6$, and $r = 10$. The shadow-line envelopes represent the spread of reconstructed responses over the full time series. Insets summarize relative errors in peak flow (Q_p), time to peak (T_p), and event volume (Vol).

3.2. Node-Level Reconstruction Accuracy

Similar to system-level reconstruction, node-level reconstruction also generally increased with sensor counts. As shown in Fig. 6, the node-level NSE distribution for both the observed events and the design event shifted upward overall as r increased. As sensor counts increased from $r = 1$ to 6 and 10, the median (50th percentile) node-level NSE increased from 0.717 to 0.960 and slightly decreased to a still high value of 0.900, respectively for the observed events, and increased from 0.526 to 0.941 and 0.965 for the 200-year design event. The upper quartile (75th percentile) remained high once more than a few sensors were used; as sensor counts increased from $r = 1$ to 6 and 10, the upper quartile NSE values increased from 0.870 to 0.995 and remained nearly unchanged at 0.995 for the observed events, and increased from 0.823 to 0.993 and 0.998 for the 200-year design event.

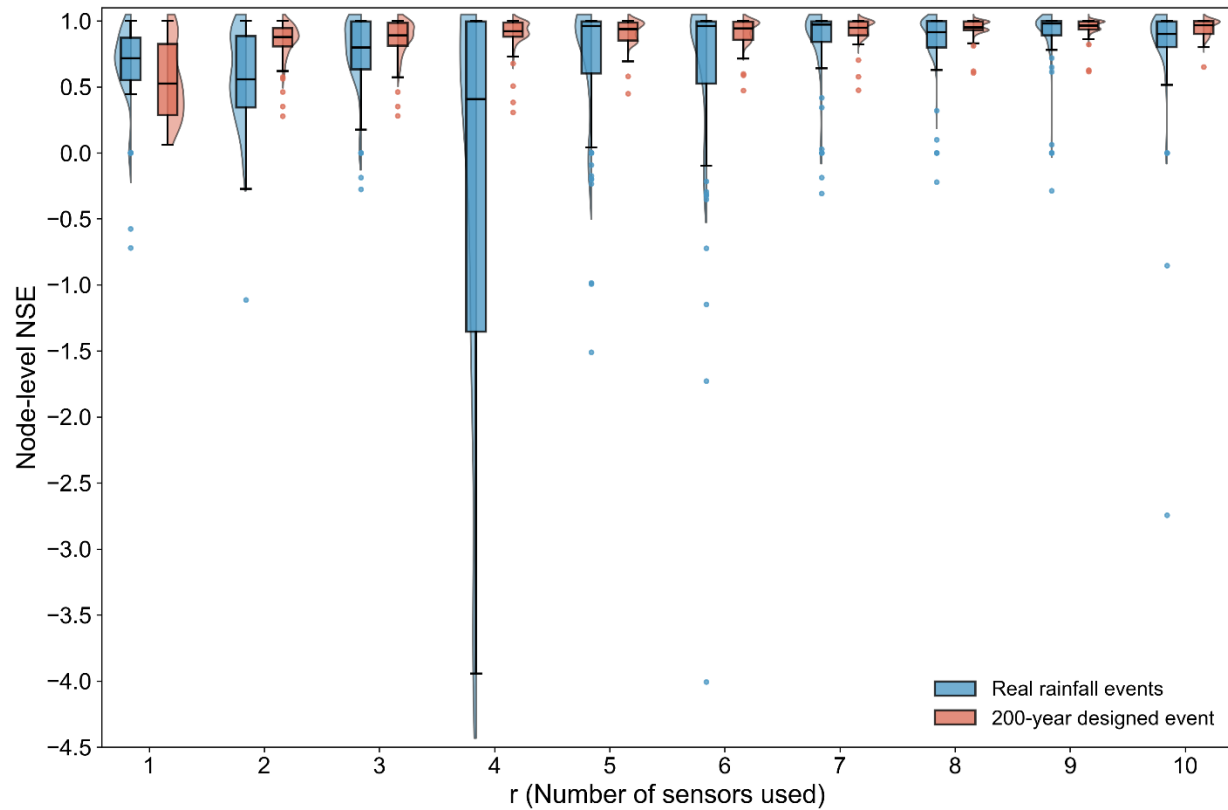


Figure 6. Node-level NSE distributions across sensor counts. Boxplots show the distribution of node-level NSE for the observed rainfall events (blue) and the 200-year designed event (red) at each sensor count r . The adjacent half-violin plots show the corresponding kernel density distributions within the 5th-95th percentile range, providing additional information on the shape

and spread of node-level reconstruction performance. The figure summarizes the variation in node-level reconstruction performance as the number of selected sensors increases from $r = 1$ to $r = 10$.

However, compared to system-level reconstruction, node-level reconstruction exhibited greater variations for the observed events. Specifically, the first quartile (25th percentile) decreased from 0.552 at $r = 1$ to 0.343 at $r = 2$ and -1.354 at $r = 4$, then recovered to 0.891 at $r = 9$ (Fig. 6). Looking at this through a different lens, 14.474% of nodes had NSE below 0.500 at $r = 1$, and this fraction increased to 52.632% at $r = 4$. Even at $r = 10$, a non-negligible fraction remained with 6.579% of nodes below 0.500 and 2.632% below 0. The 200-year design event showed a more stable upward trend; the first quartile NSE values increased from 0.285 at $r = 1$ to 0.856 at $r = 6$ and 0.901 at $r = 10$, and no nodes remained below 0.500 when $r \geq 8$. This observation indicates that system-oriented sensor placement improved local fidelity for many nodes but did not produce spatially uniform gains across the full network. The detailed spatial distribution of node-level flow reconstruction performance across nodes can be found in Figure S2–S5 in the supplementary material.

3.3. Benchmark Flow Reconstruction Performance Against Reference Placements

DSS-obtained sensor configurations produced similar reconstruction performance as that obtained by the exhaustive search with sensor counts $r = 1$ –4. The system-level NSE of exhaustive optimum remained slightly superior to DSS, but the gap was marginal (Fig. 7). The mean system-level NSEs achieved by DSS were 0.887, 0.923, 0.954, and 0.967 for $r = 1, 2, 3$, and 4, respectively, whereas the corresponding exhaustive optima were 0.890, 0.935, 0.955, and 0.973 (Fig. 7a).

DSS-obtained sensor configurations consistently produced better flow reconstruction than typical random sensor placements with sensor counts $r = 5$ –10. The random benchmark distribution was highly heavy-tailed, with many random configurations producing strongly negative system-level NSE values. In contrast, the DSS solutions remained consistently high-performing, with mean system-level NSEs ranging between 0.983–0.995 for $r = 5$ –10 (Fig. 7b).

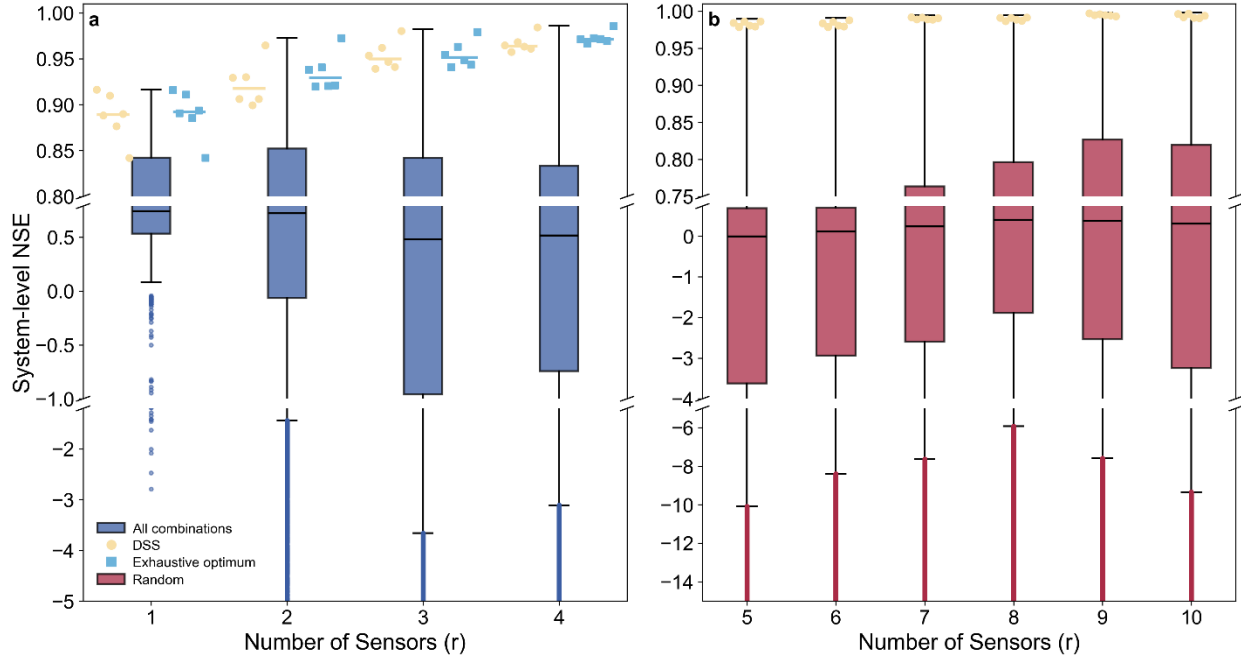


Figure 7. Benchmark comparison between DSS-selected and reference sensor sets. Panel (a) compares DSS with the exhaustive optimum for $r = 1-4$. Blue boxplots show the distribution of system-level NSE over all possible sensor combinations, yellow circles denote the DSS-selected configuration, and blue squares denote the exhaustive optimum. Panel (b) compares DSS with a random Monte Carlo benchmark for $r = 5-10$. Red boxplots show the distribution of random sensor configurations, and yellow circles denote the DSS-selected configuration.

3.4. Noise Effects on Reconstruction Accuracy

Gaussian noise added to measured data resulted in limited reduction in system-level NSE and did not alter the overall flow reconstruction performance. For example, at $r = 1$, the median system-level NSE decreased from 0.889 under clean observations to 0.889, 0.888, and 0.887 under 5%, 10%, and 15% noise, respectively, while the interquartile range changed only slightly from 0.880–0.905 to 0.877–0.910, 0.876–0.909, and 0.874–0.907 (Fig. 8).

Relative to the gains produced by adding sensors, the impact of measurement noise propagation was modest (Fig. 8). When more sensors were selected, the reduction in system-level NSE became increasingly negligible. For example, at $r = 6$, when 5%, 10%, and 15% Gaussian noises were included, the median NSE kept at 0.981 ± 0.001 with the interquartile range keeping at 0.977–0.986 (Fig. 8).

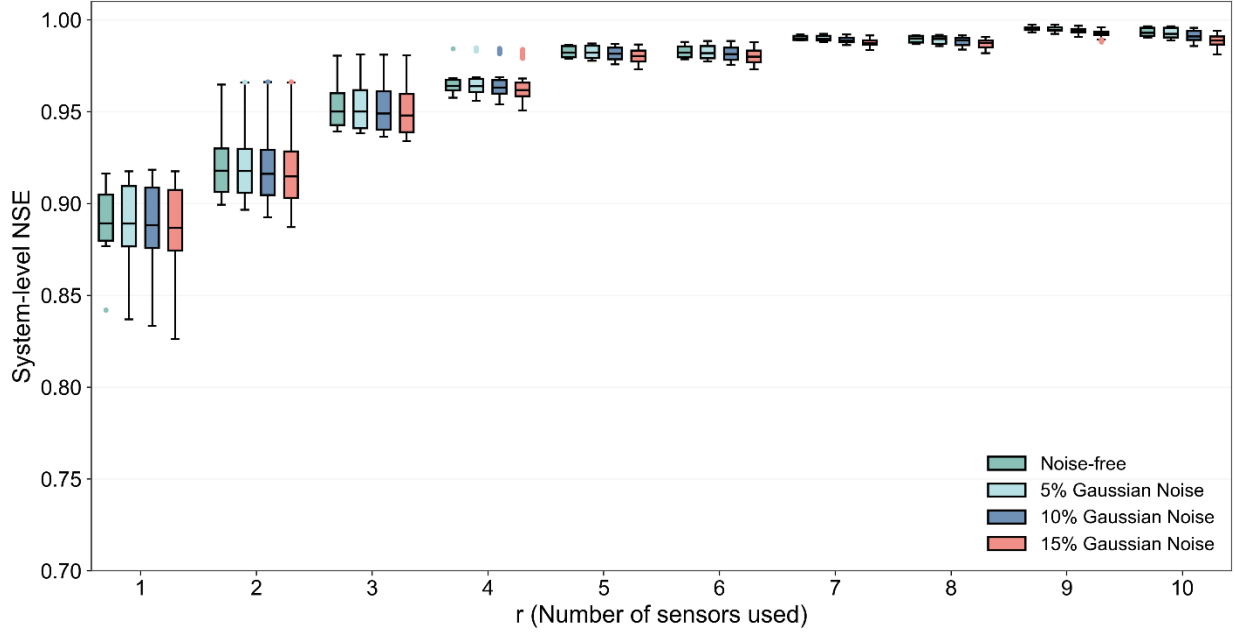


Figure 8. System-level NSE distributions under multiplicative Gaussian sensor noise. Boxplots show the system-level NSE distributions for noise-free observations and for 5%, 10%, and 15% Gaussian noise at each sensor count r . The figure summarizes the effect of measurement noise on system-level reconstruction performance across the full sensor-count range.

3.5. System-Level Sensitivity to Single-Sensor Failure

The sensor configurations identified by QR were generally robust to single-sensor failure, although sensitivity was strongly location dependent. Delta system-level NSE is defined as the decrease in mean system-level NSE after dropout, i.e., baseline NSE minus post-dropout NSE. As shown in Fig. 9, dropout of most sensors showed limited effect on the flow reconstruction performance. With sensor counts $r = 6$, removing OF-04, 163, and 184 caused small decreases in mean system-level NSE, with delta NSE values of 0.009, 0.038, and 0.007, respectively (Fig. 9a). With $r = 10$, most single-sensor dropouts also produced limited degradation, with delta system-level mean NSE below 0.055 except for the most failure-sensitive sensors (Fig. 9b).

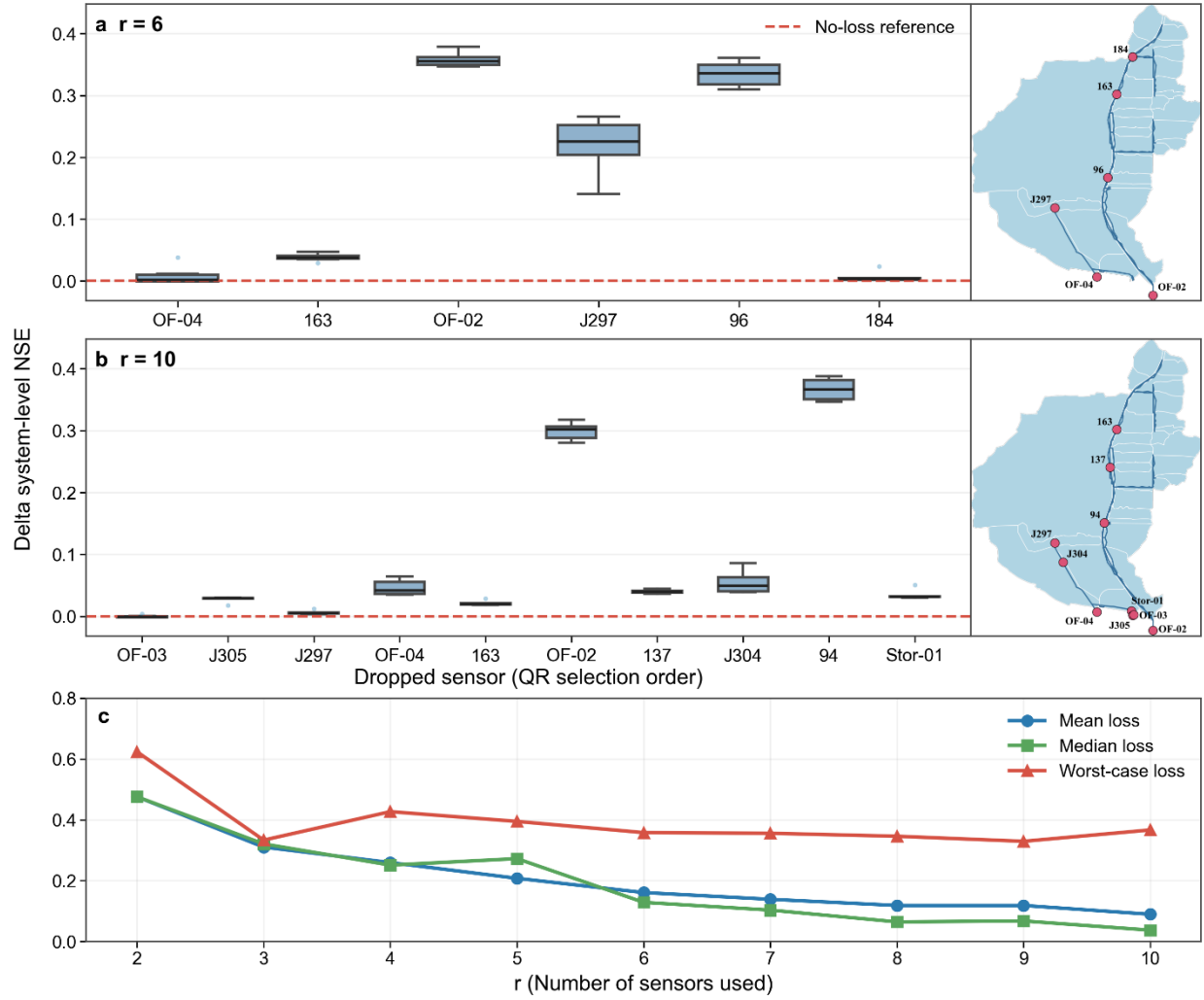


Figure 9. System-level reconstruction performance loss after single-sensor dropout at sensor counts (a) $r = 6$, (b) $r = 10$ and (c) aggregate effect of single-sensor dropout across sensor counts. In panels (a) and (b), the boxplots show the decrease in system-level NSE after removing one selected sensor from the DSS configuration, defined as the baseline mean NSE minus the post-dropout mean NSE. Sensors are ordered according to the QR selection sequence, and the dashed line denotes the no-loss reference. The maps on the right illustrate the location of the selected sensors within the network. Panel (c) summarizes the mean, median, and worst-case loss in system-level mean NSE after single-sensor dropout against sensor count.

System-level fault tolerance improved as the number of deployed sensors increased. With $r = 6$, the system-level flow reconstruction was sensitive to the failure of 3 out of 6 sensors; the dropout of OF-04, 163, and 184 had minor effects, whereas dropping OF-02, J297, and 96 reduced the

mean system-level NSE by 0.358, 0.219 and 0.335, respectively (Fig. 9a). However, with r increased to 10, sensitivity was concentrated in 2 out of 10 sensors; removing OF-02 and 94 caused delta NSE values of 0.299 and 0.367, respectively, whereas the remaining dropouts caused much smaller losses (Fig. 9b). The same pattern can be observed from Fig. 9c which shows the variation of system-level NSE loss after single-sensor dropout across sensor counts. With r increased from 2 to 6 and 10, the mean system-level NSE loss after single-sensor dropout decreased from 0.476 to 0.161 and 0.090, the median loss decreased from 0.476 to 0.128 and 0.037, and the worst-case NSE loss decreased from 0.623 to 0.358 and 0.367.

Higher sensor density improved robustness on average but did not eliminate the presence of individual failure-critical sensors; a small number of dropout cases still led to disproportionate degradation. As shown in Figure 9c, the worst-case NSE loss remained comparatively large across the full range of sensor counts. The system-level NSE showed no significant improvement after sensor count r increased above 6. With $r = 10$, dropping one critical sensor can still cause the system-level NSE to decrease by 0.367. It should be noted that because the QR sensor set is re-identified for each sensor count using the corresponding truncated modal subspace, both the selected sensor combination and the pivot order vary with r . Dropout sensitivity should therefore be interpreted as a configuration-dependent property of the sensing set at each sensor count, rather than as a one-to-one comparison of the same ordered sensors across different r . As shown in the Supplementary heatmap analyses, some sensors recur repeatedly among the most damaging dropout cases across multiple sensor-count settings, suggesting that dropout sensitivity is concentrated in a limited subset of structurally important sensing locations.

3.6. Robustness to SWMM Parameter Uncertainty

The system-level flow reconstruction performance was broadly robust across the SWMM ensembles, particularly at larger sensor counts. With sensor counts r increased from 4 to 6, 8 and 10, the mean system-level NSE obtained based on the unmodified reference SWMM model increased from 0.925 to 0.955, 0.960 and 0.974, respectively, and those obtained based on the other nine modified SWMM models increased from 0.908 to 0.945, 0.962 and 0.974, respectively (Fig. 10a). With fewer sensors (e.g., $r = 4$ to 6), the reconstruction based on the unmodified reference SWMM model tended to outperform the other modified SWMM models; whereas at larger sensor counts (e.g., $r = 8$ to 10), all SWMM model variants showed similar reconstruction performance

(Fig. 10a), demonstrating that the sparse sensing flow reconstruction approach is robust and is not an artifact of one homogenized SWMM setup.

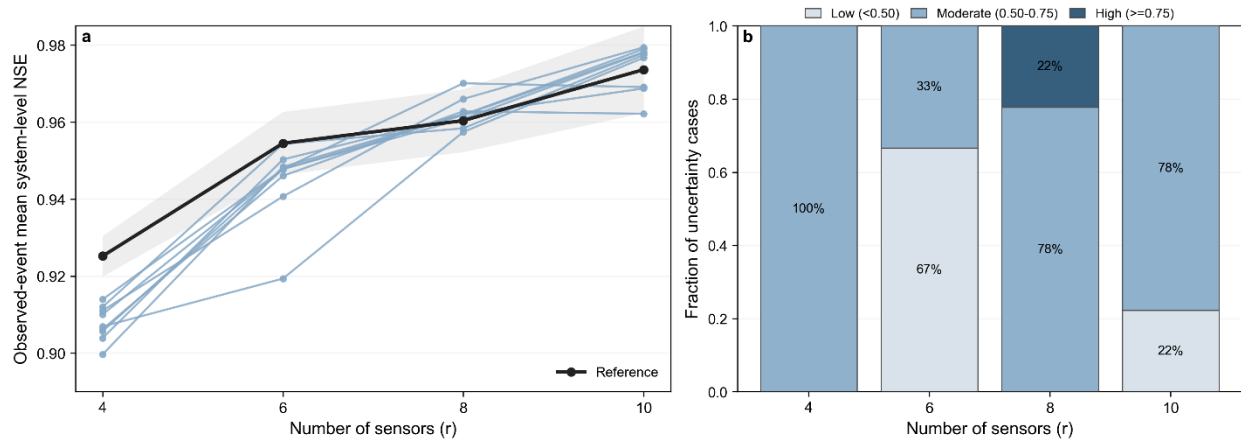


Figure 10. Robustness of system-level reconstruction and sensor selection under SWMM parameter uncertainty. Panel (a) shows the observed-event mean system-level NSE for the 10 uncertainty-model variants at $r = 4, 6, 8$, and 10 ; the shaded band denotes the event-to-event variability of the reference model across the five observed testing events. Panel (b) shows the fraction of perturbed SWMM cases falling into low (<0.50), moderate ($0.50\text{--}0.75$), and high (≥ 0.75) sensor-set Jaccard overlap categories relative to the reference-model sensor set. Detailed parameter settings and the Jaccard overlap definition are reported in Text S1 and Table S1 in the supplementary material.

Despite the robustness of the approach, the QR-obtained optimal sensors showed observable variations with the SWMM parameterizations. Relative to the reference model, all modified cases showed moderate sensor-selection overlap at $r = 4$. At $r = 6$, the overlap became more variable, with 67% of the modified cases falling into the low-overlap class (<0.50) and 33% falling into the moderate-overlap class ($0.50\text{--}0.75$). At $r = 8$, the selections became more consistent, with 78% of the cases showing moderate overlap and 22% showing high overlap (≥ 0.75). At $r = 10$, most cases remained in the moderate-overlap class (78%), although 22% still showed low overlap. These results demonstrate that while the specific locations of optimal sensors are sensitive to hydraulic parameterization, the reconstruction effectiveness of the DSS framework remains consistently high across diverse physical scenarios. Detailed parameter multipliers and the full uncertainty-case definitions are provided in Text S1 and Table S1 in the supplementary material.

4. Discussion

4.1. Sensor Placement Optimization in Urban Drainage Networks

Accurate reconstruction and prediction of system states under sparse observation conditions has long been a hot topic and a major challenge in hydrologic and environmental monitoring (An et al., 2017; Haddad, 2025; Luo et al., 2023; Qiu et al., 2023; H. Wang et al., 2024). Sensor placement optimization is thus critical to reduce the deployment and maintenance costs and to ensure that the limited observations are maximally informative for reconstructing or predicting spatially distributed variables and supporting timely decision-making.

Prior studies have shown that 5–15% well-placed sensors can effectively predict flow conditions and detect potential risk in urban drainage networks using advanced data analytics. For instance, Farahmand et al. (2022) proposed a comprehensive network observability framework and applied it to a drainage network in Harris County, Texas. With this method, they achieved efficient flood monitoring coverage by monitoring the top 10% of critical nodes using only 121 sensors out of 455. Grimaldi et al. (2024) showed that deploying sensors at just 4 key cross-sections based on feature importance assessment combined with ML could match the flood prediction accuracy of full-network monitoring across 26 sites under a 6-hour warning window. Huang et al. (2025) applied a Bayesian decision theory-based approach to optimize sensor placement, maintaining average water level prediction error to 0.048 m with only 5 sensors in a 54-node network. Wang et al. (2023) proposed a re-clustering optimization method and used information entropy to quantify monitoring effectiveness, attaining maximal coverage with 28–38 sensors in a 786-node network.

Recently, Zheng et al. (2025) conducted a similar work as our study. They introduced a multi-objective framework that integrates information theory and matrix completion to optimize and evaluate urban drainage sensor networks. They used Value of Information (VOI) and Transinformation Entropy (TE) to identify high-value sensor locations and applied Non-negative Matrix Factorization (NMF) to evaluate the performance of the sensor network based on reconstruction accuracy. Applied to a drainage network with 878 nodes, their approach achieved flooding risk assessment accuracy of 76% and 82% using 4 and 8 sensors, respectively, with corresponding perception errors of 33% and 29%. By contrast, DSS directly targets reduced-order reconstruction of dynamic system behavior. In the present 77-node network, DSS achieved a mean system-level NSE of 0.914 with two sensors and 0.949 with three sensors across the five observed

events, while also remaining robust to moderate measurement noise. Such a framework can complement traditional localized or discrete point-based monitoring and is particularly valuable for flashy urban drainage systems, where reconstructing continuous hydraulic transients and peak timing is critical.

4.2. Factors Driving Location-Dependent Sensitivity to Sensor Failure

As observed in this study (section 3.5), while QR rank identifies the primary sensors intended to maximize information gain, rank alone was an insufficient predictor of failure impact. System-level flow reconstruction performance showed higher sensitivity to the failure of certain sensors, and this sensitivity varied with sensor counts. This behavior is consistent with prior studies showing that monitoring-network performance under failure depends more on redundancy, information uniqueness, and post-failure functionality than on the originally optimized sensor layout. Specifically, Jung and Kim (2017) showed that the failure of a key meter can sharply reduce detection performance when other meters only provide supplementary information. Zhang et al. (2020) ranked sensors by their ability to maintain monitoring resilience during failure. In sewer and urban drainage systems, information-theoretic placement studies indicate that sensor value depends on both information content and redundancy, often quantified by metrics such as joint entropy, total correlation, mutual information, value of information, and transinformation entropy (Banik et al., 2017; Crowley et al., 2025; Zheng et al., 2025).

Based on existing literature, the sensitivity of a structural monitoring system to sensor failure is a multidimensional problem, and a sensor's true criticality is a product of its unique role in maintaining system stability and its modal contribution, which may not align perfectly with its selection order. Here we selected three mathematical metrics, including relative projection residual (RPR) of the failed sensor (Eq. 9), condition number (K) after sensor dropout (Eq. 10), and maximum modal loading (L) of the failed sensor in the retained basis (Eq. 11), and evaluated their relative impact on the system-level flow reconstruction performance after sensor failure in addition to QR rank. RPR measures the relative projection error after removing one sensor, which represents redundancy; larger RPR values indicate lower redundancy and lower replaceability. K measures the numerical stability after sensor dropout; larger K values indicate increased numerical instability and higher sensitivity to missing states. L measures the linear statistical significance of

the failed sensor on flow reconstruction; larger L values indicate stronger loading on at least one dominant mode.

$$RPR_i = \frac{\|u_i - \Pi_{U_{-i}}(u_i)\|_2}{\|u_i\|_2} \quad (9)$$

where u_i is the row of the truncated basis corresponding to the failed sensor, U_{-i} is the matrix formed by the remaining selected sensor rows, $\Pi_{U_{-i}}(u_i)$ is the projection of u_i onto the row space of the remaining sensors and RPR_i is the relative projection error after removing sensor i .

$$\kappa_i = \frac{\sigma_{\max}(\Psi_{-i,r})}{\sigma_{\min}(\Psi_{-i,r})} \quad (10)$$

where $\Psi_{-i,r}$ is the truncated sensing matrix after removing sensor i , $\sigma_{\max}$ is the largest singular value of the post-dropout sensing matrix, $\sigma_{\min}$ is the smallest singular value of the post-dropout sensing matrix and κ_i is the condition number measuring numerical stability after dropout.

$$L_i = \max_{1 \leq j \leq r} |\Psi_r(i, j)| \quad (11)$$

where $\Psi_r(i, j)$ is the entry of the failed sensor row in retained mode j and L_i is the largest absolute modal coefficient across the retained modes.

More specifically, we first fitted a QR-rank-only ordinary least squares (OLS) regression model to quantify the fraction of the observed system-level dropout loss that could be explained by the QR selection order alone. Additional explanatory variables, including the RPR, K , and L were then incorporated sequentially to evaluate whether sensor dropout sensitivity was associated with factors beyond QR rank. This analysis was intended as a post-hoc explanatory test rather than a transferable predictive model; therefore, the resulting R^2 values were interpreted as in-sample explanatory power rather than out-of-sample predictive skill.

The multivariable analysis results (Fig. 11a) indicated that a single factor alone does not fully explain the observed variability in sensor sensitivity. The QR-rank-only model achieved an adjusted R^2 of only 0.159, indicating limited explanatory power for the observed dropout-loss variability. After sequentially incorporating RPR, K , and L , the adjusted R^2 increased to 0.509, 0.585, and 0.694, respectively. The same increasing pattern was observed for leave-one-out cross-validation (R^2), which increased from 0.111 to 0.479, 0.557, and 0.664. This consistency indicates

that the improvement was not solely a mathematical artifact of model expansion. The bootstrap leave-one-variable-out analysis (Fig. 11b) further supported this interpretation: L contributed the largest share of the summed median ΔR^2 (64%), followed by QR selection order (22%), K (7%), and RPR (6%). Figures S6–S9 in the supplementary material illustrate the results across sensor counts.

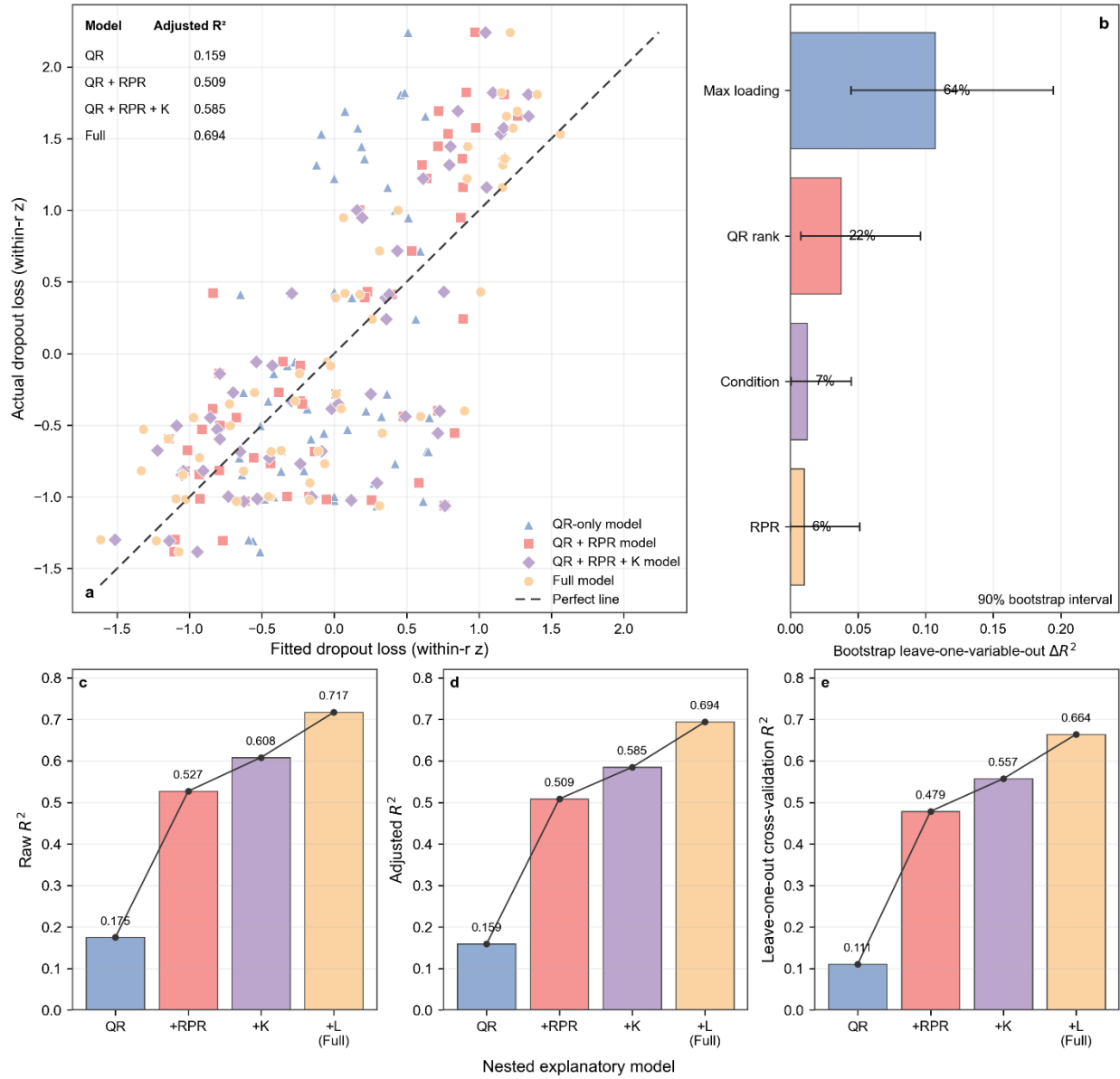


Figure 11. Structural factors associated with heterogeneous dropout sensitivity. Panel (a) compares actual and fitted within-r standardized system-level dropout losses from nested explanatory ordinary least-squares models with progressively added variables: QR rank, RPR, K, and L. The

dashed line indicates the 1:1 reference, and the inset table reports the adjusted R^2 of each nested model. Panel (b) shows bootstrap leave-one-variable-out ΔR^2 contributions for the full model, with bars representing median ΔR^2 contributions and whiskers representing the 90% bootstrap interval, defined by the 5th and 95th percentiles. Percentages denote each variable's relative share of the summed median ΔR^2 . Panels (c)–(e) summarize raw R^2 , adjusted R^2 , and leave-one-out cross-validation R^2 , respectively, for the same nested model sequence.

Taken together, these findings support a joint-mechanism interpretation of dropout vulnerability. Single-sensor failure was most damaging when the removed sensor had low redundancy with the remaining sensors, contributed strongly to the retained modal structure, and its removal produced a poorly conditioned post-dropout sensing matrix. Robust sensing design should therefore be evaluated not only by average reconstruction accuracy, but also by how redundancy, conditioning, and modal importance are distributed within the selected sensor set.

4.3. Limitations and Future Work

Despite its demonstrated strengths, the current approach has several limitations that warrant further exploration. First, without long-term field measurements, the study relies on simulated flow data generated by physics-based models such as EPA-SWMM to construct the sensing basis. Consequently, despite the robustness of our approach to model parameterization as illustrated, any inaccuracies in model structure, parameter calibration, or rainfall input will affect the model output (Høybye & Rosbjerg, 1999; Sharif et al., 2004) and propagate into the selection of sensor locations and impact reconstruction performance. Improving model fidelity through integration with field measurements, especially from long-term monitoring campaigns, is essential to enhance the applicability and transferability of the DSS approach.

Second, DSS is fundamentally a linear reduced-order reconstruction framework, which limits its ability to represent strongly nonlinear and threshold-driven node behavior. This limitation is most evident at overflow-related nodes that remain at zero under most conditions but become active only when hydraulic thresholds are exceeded during intense rainfall. For such nodes, a linear modal reconstruction can generate spurious weak flows or miss abrupt activation, leading to false positives or false negatives at the node level even when system-level performance remains high.

Future work should therefore explore regime-switching, nonlinear manifold, or hybrid physics-informed learning frameworks to better represent threshold-controlled drainage responses.

Third, the current algorithm identifies the optimal sensor placement using a single variable – flowrate – and only enables its reconstruction. However, many actual monitoring applications involve multi-variable sensing, such as flowrate, water level, and water quality parameters (Alam et al., 2021; Nygaard, 2006; Sebicho et al., 2024; R. Zhang, Wang, et al., 2023). Extending DSS to handle multiple target variables will improve the representativeness of the optimal locations identified and enhance its applicability across water systems.

Fourth, the framework is currently applied to reconstruction tasks based on existing or incoming sensor data and lacks forecasting capacity. Future work should explore predictive extensions by integrating DSS with data-driven forecasting models, such as long short-term memory networks (LSTM) or physics-informed machine learning (PIML). This hybridization could enable real-time system state estimation and short-term flood forecasting with minimal sensors. The framework can be further coupled with real-time control (RTC) strategies to enhance adaptive urban flood management.

5. Conclusion

This study presents a data-model integrated framework that combines process-based simulation (via EPA-SWMM) with data-driven sparse sensing (DSS) to optimize sensor placement for urban flood reconstruction under data-scarce conditions. Using the Woodland catchment in Duluth, MN as a case study, we demonstrated that a small number of strategically selected monitoring nodes, identified through singular value decomposition (SVD) of simulation data and QR factorization with column pivoting, can effectively capture the dominant hydrodynamic patterns in a stormwater network and reconstruct flowrate conditions at the system level.

Despite variations in reconstruction performance at the individual node level, the system-level results remained consistently high. For the five observed events, the mean system-level NSE increased from 0.896 at $r = 1$ to 0.993 at $r = 10$, while the 200-year design event improved from 0.842 to 0.994. With three sensors, the mean system-level NSE already reached 0.949 across the observed events, and eight sensors increased it to 0.989. Benchmark tests kept DSS close to the

exhaustive optimum for sensor counts $r = 1-4$, with mean gaps of 0.002–0.013, and that DSS consistently outperformed typical random placements for $r = 5-10$.

The framework remained robust to moderate multiplicative Gaussian noise and single sensor failure. Sensor-dropout analyses showed that average vulnerability decreased with sensor density but remained concentrated in a small subset of structurally critical sensors. Taken together, these results support DSS as an efficient and interpretable strategy for system-level urban drainage monitoring under constrained sensing. Future development should focus on reducing dependence on simulation uncertainty, improving the accuracy of threshold-controlled nodes (e.g., overflows) that exhibit sudden, non-linear hydraulic responses, and integrating reconstruction with predictive forecasting and control models.

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Data Availability Statement

All the data and scripts used in this study can be accessed from the GitHub repository (<https://github.com/DzhZzz/Data-Driven-Sparse-Sensing.git>).

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Author Contribution

Zihang Ding: Conceptualization; Data Curation; Formal analysis; Investigation; Methodology; Validation; Visualization; Writing – original draft; Writing – review & editing. Imran Md. Azizul Islam: Validation, Writing – review & editing. Amit Kumar: Validation, Writing – review & editing. Mila Avellar Montezuma: review & editing; Ruihang Zhang: Validation, Writing – review & editing. Kun Zhang: Conceptualization; Funding acquisition; Methodology; Project administration; Resources; Software; Supervision; Validation; Writing – review & editing.

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